

CHAPTER (3)

SOLAR PHOTOVOLTAIC SYSTEM DESIGN

3.1 Introduction

With the increasing global demand for clean and sustainable energy sources, photovoltaic (PV) systems have become one of the most effective solutions for electricity generation. Solar energy offers an environmentally friendly alternative to conventional energy resources by converting sunlight directly into electrical power without producing harmful emissions.

This chapter presents the design and operation principles of the proposed 150 kW AC grid-connected solar photovoltaic system. The chapter includes the fundamentals of solar energy conversion, system configuration, component selection, and the methodology used for system sizing and performance evaluation.

3.1.1 Definition of Solar Energy

Solar energy is a renewable energy source generated from solar radiation emitted by the sun. Photovoltaic technology converts this solar radiation directly into electrical energy using semiconductor materials.

Photovoltaic modules are composed of multiple solar cells connected electrically to generate useful voltage and current levels. These modules capture solar radiation and transform it into direct current (DC) electricity, which is later converted into alternating current (AC) using grid-connected inverters.

The use of solar energy contributes to reducing greenhouse gas emissions, decreasing dependence on fossil fuels, and improving energy sustainability.

3.1.2 Role of Solar Energy in Grid-Connected Power Systems

Grid-connected photovoltaic systems are increasingly used in commercial and institutional buildings due to their high reliability and economic benefits.

In this project, solar energy is utilized to generate electrical power through rooftop-mounted photovoltaic arrays connected directly to the utility grid through string inverters.

The generated energy contributes to reducing conventional electricity consumption while improving overall energy efficiency and supporting sustainable development goals.

3.2 Overview of Photovoltaic (PV) Systems

The sun is considered the largest renewable energy source available to Earth. Solar photovoltaic systems convert solar radiation directly into electrical power using semiconductor-based photovoltaic cells.

A photovoltaic system typically consists of:

- PV Modules (Solar Panels)
- Mounting Structure
- DC Cabling
- Inverters
- Protection Devices
- AC Distribution System
- Utility Grid Connection

The performance of a PV system depends on several factors including solar irradiation, ambient temperature, module efficiency, orientation angle, and electrical losses.

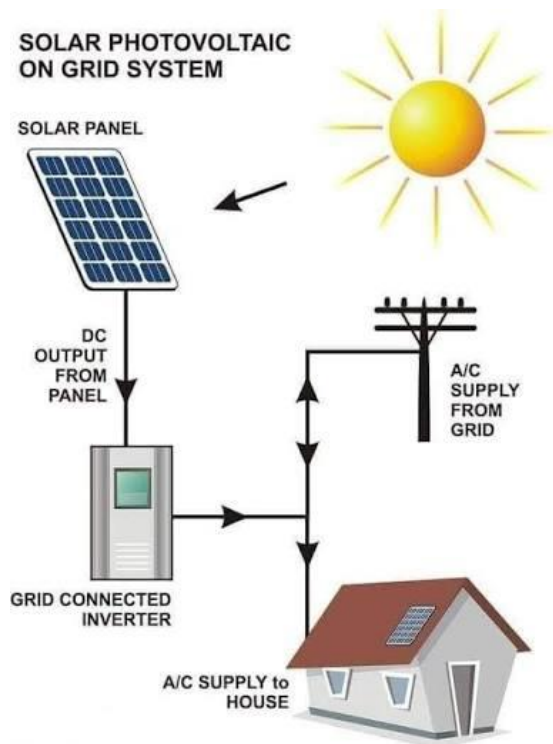


Figure (3.1): General Configuration of Grid-Connected PV System

3.2.1 Principle of Electricity Generation from Solar Energy

Photovoltaic electricity generation is based on the photovoltaic effect.

When solar radiation reaches the surface of a photovoltaic cell, photons transfer their energy to electrons inside the semiconductor material.

As a result, electrons gain sufficient energy to move and generate an electric current.

The internal electric field inside the solar cell forces these electrons to move in one direction, producing direct current (DC).

This DC power is collected from multiple PV modules and delivered to an inverter where it is converted into alternating current (AC) suitable for supplying electrical loads and exporting energy to the utility grid.

The overall conversion process can be summarized as:

Solar Radiation



PV Modules



DC Electricity



Inverter



AC Electricity



Building / Utility Grid

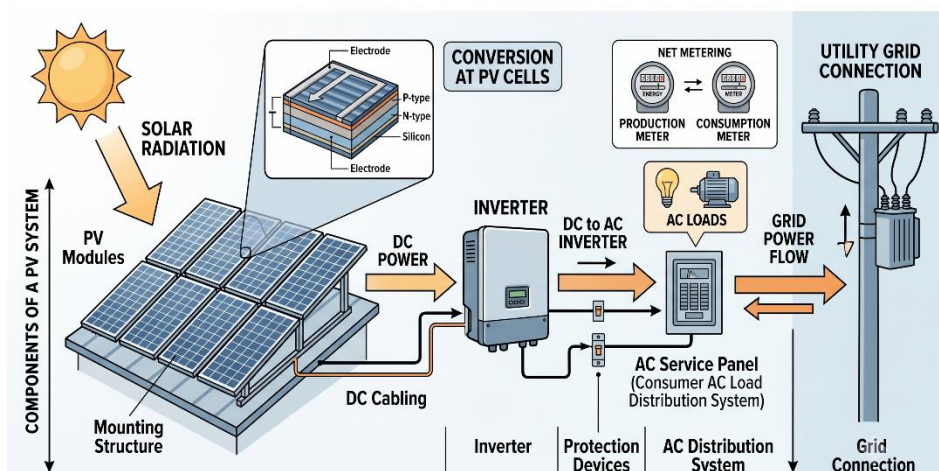


Figure (3.2): Photovoltaic Energy Conversion Process

3.2.2 Advantages of Solar Energy

Solar energy is considered one of the most promising renewable energy sources due to its environmental, technical, and economic benefits. In grid-connected photovoltaic applications, solar energy provides an effective solution for sustainable electricity generation.

The main advantages of solar energy include:

1. **Renewable and Sustainable Source**
Solar energy is continuously available through solar radiation and does not depend on finite fuel reserves.
2. **Environmentally Friendly Energy Production**
Photovoltaic systems generate electricity without emitting greenhouse gases or air pollutants during operation.
3. **Reduced Electricity Consumption from the Utility Grid**
Grid-connected PV systems decrease dependence on conventional electrical generation and reduce operational energy costs.
4. **Low Operating and Maintenance Requirements**
Solar PV systems have relatively low maintenance requirements after installation and offer long operational lifetimes.
5. **High Suitability for Rooftop Applications**
PV systems can efficiently utilize available rooftop areas without requiring additional land.
6. **Long System Lifetime**
Modern PV systems typically operate efficiently for more than 25 years with gradual performance degradation.

3.2.3 Limitations of Solar Energy

Despite its advantages, photovoltaic systems have several technical and operational limitations that must be considered during system design.

1. Dependence on Solar Irradiation

PV output varies according to weather conditions, cloud cover, and available sunlight.

2. Reduced Efficiency at High Temperature

Solar module performance decreases as operating temperature increases.

3. Initial Installation Cost

The initial investment for PV systems may be relatively high compared with conventional electrical installations.

4. Space Requirement

Large installed capacities require sufficient roof or land area for module installation and maintenance access.

5. System Losses

Actual energy production is affected by losses due to temperature, wiring, inverter conversion, mismatch, and soiling.

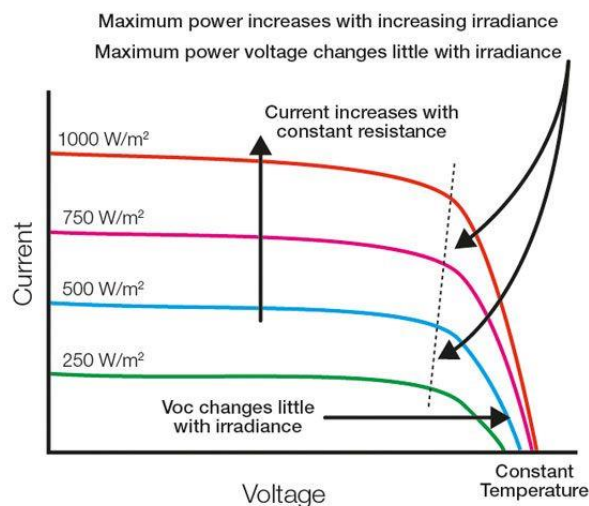


Figure (3.3): Factors Affecting PV System Performance

3.2.4 Components of the Proposed PV System

A grid-connected photovoltaic system consists of several integrated electrical and mechanical components that work together to convert solar radiation into usable electrical power.

The proposed 150 kW AC PV system consists of:

1. Photovoltaic Modules (PV Modules)

PV modules convert solar radiation into direct current (DC) electricity.

- **Selected Module: Suntech Ultra V STP555S-C72/Vmh**
- **Module Rating: 555 W**
- **Total Number of Modules: 320 Modules**
- **Total Installed Capacity: 177.6 kWp**

2. Mounting Structure

The mounting structure supports the PV modules and provides the required tilt angle and mechanical stability.

- **Selected Configuration: Fixed Tilt Structure**
- **Tilt Angle: 25°**

3. String Inverters

Inverters convert DC electricity generated by PV modules into AC electricity compatible with the utility grid.

- **Selected Inverter: Growatt MAX 75KTL3 LV**
- **Quantity: 2 Units**
- **Rated Output Power: 150 kW AC**

4. Electrical Protection System

Includes DC and AC protection devices such as:

- **DC Fuse : 20 A**
- **Circuit Breakers (MCCB): 160 A**

3.2.5 Types of Photovoltaic Systems

Photovoltaic systems can be classified according to their connection method and operational mode.

1. Grid-Connected PV System (Selected System)

This type of system operates while connected directly to the electrical utility network.

Characteristics:

- No battery storage required
- Higher overall efficiency
- Reduced installation cost
- Excess generated energy can be exported to the grid
- Suitable for commercial and institutional buildings

This project adopts a Grid-Connected configuration due to its economic and operational advantages.

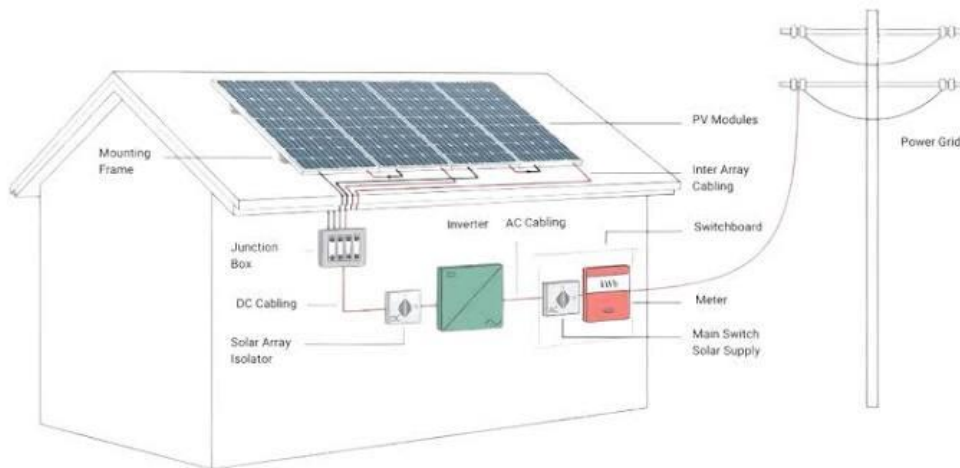


Figure (3.4): Grid-Connected Solar PV System

2. Off-Grid PV System

Off-grid systems operate independently from the utility grid and require battery storage systems.

Characteristics:

- Suitable for remote locations
- Requires energy storage
- Higher maintenance requirements
- Higher installation cost

Since the project location has utility grid availability, off-grid configuration was not selected.

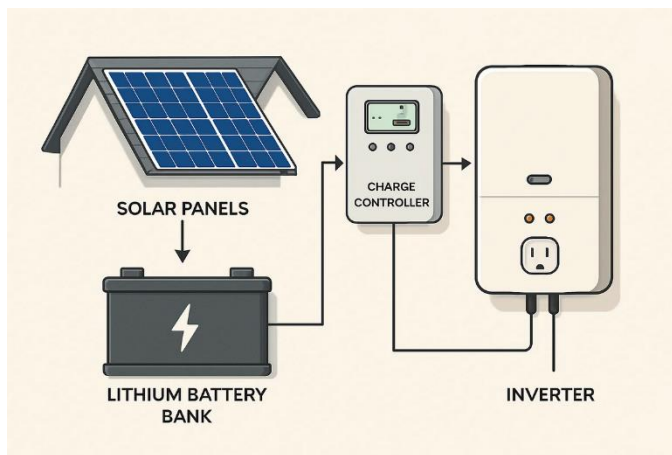


Figure (3.5): Off-Grid Solar PV System

3.2.5 Benefits of Photovoltaic Systems

Photovoltaic (PV) systems have become one of the most widely adopted renewable energy technologies due to their environmental, operational, and economic advantages. Their implementation in commercial and institutional buildings contributes significantly to energy sustainability.

The major benefits of photovoltaic systems include:

1. **High Reliability and Low Maintenance Requirements**

PV systems contain very few moving components, which reduces mechanical failures and simplifies maintenance activities. Periodic visual inspection and cleaning are generally sufficient to maintain system performance.

2. **Long Operational Lifetime**

Modern photovoltaic modules are designed for long-term operation and typically maintain acceptable performance levels for more than 25 years.

3. **Environmentally Friendly Operation**

PV systems generate electricity without combustion processes and therefore produce no direct greenhouse gas emissions or atmospheric pollutants during operation.

4. **Reduction of Utility Electricity Consumption**

Grid-connected PV systems reduce dependence on conventional electrical generation and lower building energy consumption from the utility network.

5. **Modular and Scalable Design**

PV systems are highly flexible and can be expanded or modified according to future load growth and available installation area.

6. **Efficient Utilization of Rooftop Areas**

Rooftop PV installations allow electricity generation without requiring additional land resources.

3.2.6 Limitations of Solar Photovoltaic Systems

Although photovoltaic systems provide significant advantages, several technical limitations must be considered during system design and operation.

1. Dependence on Solar Availability

Electricity generation depends on available solar irradiance and varies according to weather conditions and daylight duration.

2. Performance Reduction at High Temperatures

Solar module efficiency decreases as operating temperature increases.

3. Initial Installation Cost

The initial investment cost of PV modules, mounting systems, inverters, and electrical infrastructure may be relatively high.

4. Large Installation Area Requirement

Higher installed capacities require sufficient roof area to accommodate module placement and maintenance pathways.

5. Conversion Losses

Losses occur throughout the system due to module temperature effects, wiring resistance, mismatch, inverter conversion, and environmental conditions.

6. Dependence on Grid Availability

Grid-connected systems require utility grid presence for synchronization and operation.

3.3 Fundamentals and Basic Concepts

To fully understand the proposed photovoltaic system design and performance, several fundamental concepts related to photovoltaic energy conversion must be introduced.

This section presents the operating principles of solar cells, photovoltaic characteristics, and maximum power extraction concepts used in the proposed system.

3.3.1 Solar Cell

A solar cell is the basic energy conversion element in a photovoltaic system.

Solar cells are semiconductor devices that directly convert solar radiation into electrical energy through the photovoltaic effect.

Multiple cells are electrically interconnected to form photovoltaic modules capable of generating practical voltage and power levels.

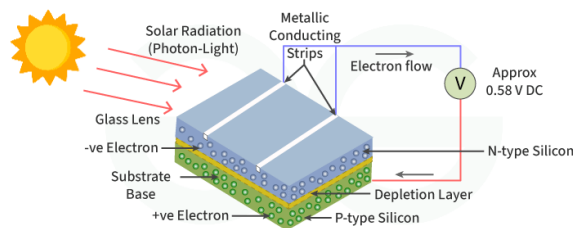


Figure (3.6): Structure of a Solar Cell

3.3.2 Solar Cell Structure

A photovoltaic cell is commonly manufactured using silicon semiconductor material and consists of two electrically treated regions forming a PN junction.

N-Type Layer

This layer contains additional free electrons introduced through controlled doping and is positioned toward incoming solar radiation.

P-Type Layer

This layer contains electron vacancies (holes), creating the electric field required for charge separation.

When these layers are combined, an internal electric field is formed that enables electricity generation.

3.3.3 Solar Cell Operating Principle

The photovoltaic effect occurs when solar radiation reaches the semiconductor surface.

Photons transfer their energy to electrons inside the material.

Once electrons gain sufficient energy, they move through the internal electric field of the cell and generate direct current (DC).

The generated DC electricity is collected and supplied to the inverter where it is converted into alternating current (AC).

The conversion process can be summarized as:

Solar Radiation

↓

Solar Cell

↓

DC Electricity

↓

Inverter

↓

AC Electricity

3.3.4 Solar Cell Electrical Characteristics

The electrical behavior of photovoltaic cells is represented using:

- Current–Voltage (I–V) Curve
- Power–Voltage (P–V) Curve

Important operating parameters include:

Short Circuit Current (I_{sc})

Current generated when output voltage equals zero.

Open Circuit Voltage (V_{oc})

Maximum voltage generated under open circuit conditions.

Maximum Power Point (MPP)

Operating point where the product of voltage and current reaches its maximum value.

Maximum Power:

$$P_{\max} = V_{\text{mpp}} \times I_{\text{mpp}}$$

These characteristics change according to:

- Solar Irradiance
- Module Temperature
- Environmental Conditions

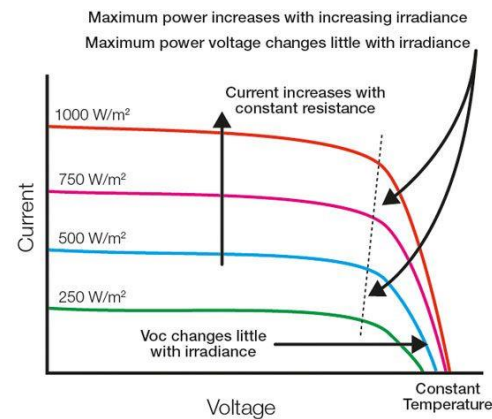
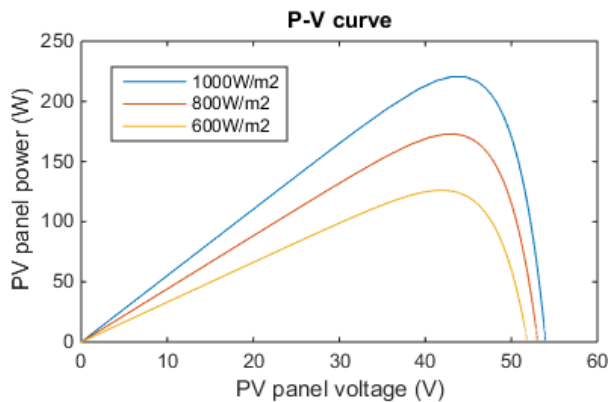


Figure (3.7,3.8): I-V and P-V Curves

3.3.5 Maximum Power Point Tracking (MPPT)

Maximum Power Point Tracking (MPPT) is an electronic control technique integrated inside modern grid-connected inverters.

The objective of MPPT is to continuously adjust the operating voltage and current of photovoltaic arrays to extract the maximum available power under varying environmental conditions.

For photovoltaic modules:

Maximum Power:

$$P_{\max} = V_{\text{mpp}} \times I_{\text{mpp}}$$

In the proposed project, MPPT functionality is implemented internally within the selected Growatt string inverters.

This allows the PV system to maintain high energy yield despite temperature variation and changing solar irradiation.

3.4 Site and Climate Data Input

3.4.1 Station Location

The first stage of photovoltaic system design in PVsyst is defining the project location and importing accurate meteorological data.

For the proposed project, the selected site is located in Egypt at:

- Latitude: 30.009° N
Longitude: 31.758° E
Altitude: 310 m
- Utilized Area for PV Installation: Approximately 1529 m²
- Annual Energy Production: 296,040 kWh/year
- Specific Production: 1852 kWh/kWp/year

Meteorological data were obtained using the PVGIS SARA3 database integrated with PVsyst to ensure realistic estimation of system performance.

The imported climate data include:

- Global Horizontal Irradiation (GHI)
- Diffuse Horizontal Irradiation (DHI)
- Ambient Temperature
- Wind Speed
- Solar Geometry Parameters

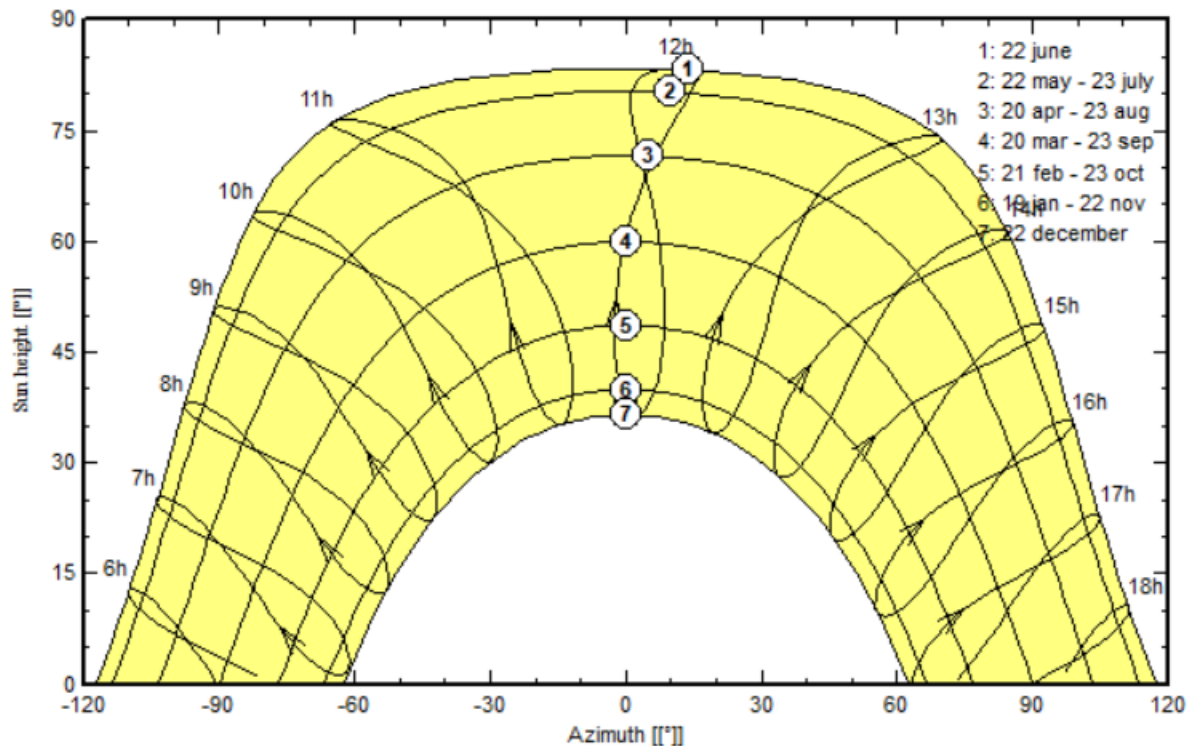
The annual in-plane irradiation was estimated at approximately: 2480.89 kWh/m²/year

These climate inputs are fundamental for evaluating annual energy generation and optimizing system orientation.

The sun-path analysis was also used to determine the optimum panel orientation and installation angle.



Solar paths at Graduation_Egypt, (Lat. 30.0°N, long. 31.8°E, alt. 310 m) - Legal Time



3.4.2 PV Module and Inverter Selection

PV Module Selection

The selected photovoltaic module is: Suntech Ultra V (STP555S-C72/Vmh)

This module was selected due to its high efficiency and suitability for rooftop commercial installations.

Main Specifications:

Rated Power (P_{max}): 555 W

Module Efficiency: 21.5%

Open Circuit Voltage (V_{oc}): 50.07 V

Short Circuit Current (I_{sc}): 14.07 A

Voltage at Maximum Power (V_{mp}): 42.24 V

Current at Maximum Power (I_{mp}): 13.14 A

The selected module provides high energy density and efficient utilization of available roof area.



Ultra V

HALF-CELL MONOFACIAL MODULE

TYPE: STPXXXS - C72/Vmh

POWER OUTPUT	MAX
535-555W	21.5%

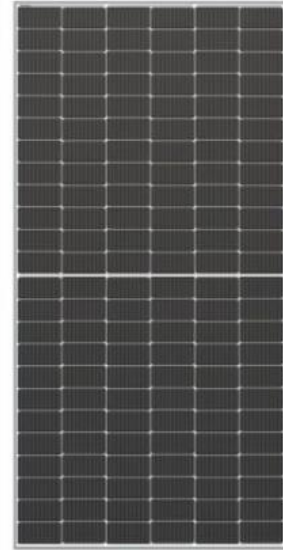


Figure (3.21): Selected PV Module

Inverter Selection

The selected inverter is: Growatt MAX 75KTL3 LV

Quantity: 2 Units

Total Installed AC Capacity: 150 kW

Main Specifications:

Nominal AC Power: 75 kW

Maximum Efficiency: 99%

Maximum DC Voltage: 1100 V

MPPT Inputs: 7 MPPT

Grid Voltage: 400 V

The inverter was selected to provide efficient DC/AC conversion while maintaining acceptable DC loading conditions.



Figure (3.22): Selected Inverter

3.4.3 System Sizing and Configuration

PVsyst software was used to optimize system sizing and evaluate the expected energy performance.

The selected system configuration is:

System Type:

Grid-Connected Rooftop PV System

Total Installed DC Capacity: 177.6 kWp

Total Installed AC Capacity: 150 kW

Total Number of Modules: 320 Modules

Module Power: 555 W

Total Inverters: 2 Units

Inverter Rating: 75 kW each

Tilt Angle: 25°

Azimuth: 0°

DC/AC Ratio: $177.6 / 150 = 1.18$

Temperature Losses $\approx 8\text{--}10\%$

DC Wiring Losses $\approx 1\%$

Mismatch Losses $\approx 1\%$

Soiling Losses $\approx 3\%$

Inverter Conversion Losses $\approx 1\text{--}2\%$

This configuration achieves efficient utilization of roof space while maintaining acceptable operating margins.

Figure (3.23): PV System Configuration in PVsyst

3.4.4 Energy Simulation and Outputs

After completing the system design, an annual performance simulation was conducted using PVsyst.

Simulation Results:

Annual Energy Production:

$\approx 336,848 \text{ kWh/year}$

Specific Production:

$\approx 1896 \text{ kWh/kWp/year}$

Estimated Average Daily Production:

$\approx 923 \text{ kWh/day}$

Performance Ratio (PR):

(To be inserted from PVsyst final report)

Estimated Monthly Production:

$\approx 28 \text{ MWh/month (average)}$

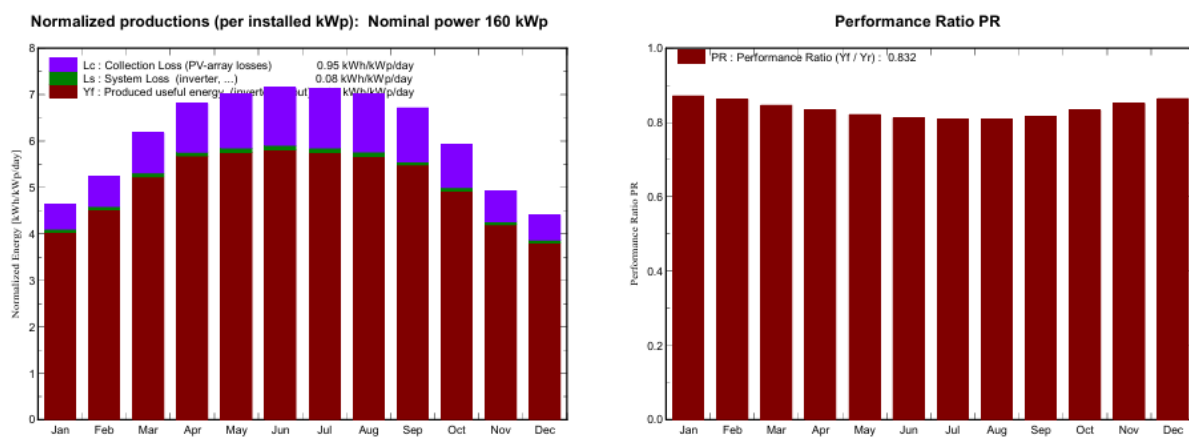
These results indicate that the designed photovoltaic system is capable of generating sufficient electrical energy to support the building load while reducing dependence on conventional electricity generation.

Main simulation results

System Production

Produced Energy **296040 kWh/year**
Performance Ratio PR **83.2 %**

Specific prod. **1852 kWh/kWp/year**



New simulation variant

Balances and main results

	GlobHor kWh/m ²	T Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	EffArrR %	EffSysR %
January	102.1	13.13	143.5	139.5	20277	19978	18.99	18.71
February	115.4	14.14	146.9	143.1	20539	20235	18.80	18.52
March	166.7	17.26	191.5	186.5	26304	25907	18.46	18.18
April	195.3	20.20	204.2	198.3	27623	27218	18.18	17.92
May	225.4	24.03	217.4	210.9	28974	28538	17.91	17.64
June	232.6	26.55	214.4	207.9	28284	27839	17.73	17.45
July	235.0	28.26	220.7	214.0	28942	28491	17.63	17.35
August	216.8	28.06	217.7	211.4	28566	28152	17.64	17.38
September	181.1	25.96	201.4	196.0	26696	26304	17.82	17.56
October	148.7	23.04	183.5	178.8	24763	24405	18.14	17.88
November	109.7	18.39	147.9	144.0	20427	20131	18.57	18.30
December	95.2	14.83	136.4	132.9	19139	18843	18.85	18.56
Year	2023.9	21.19	2225.4	2163.3	300534	296040	18.15	17.88

Legends: GlobHor Horizontal global irradiation EArray Effective energy at the output of the array
T Amb Ambient Temperature E_Grid Energy injected into grid
GlobInc Global incident in coll. plane EffArrR Effic. Eout array / rough area
GlobEff Effective Global, corr. for IAM and shadings EffSysR Effic. Eout system / rough area

Figure (3.22): Energy simulation and outputs.

3.4.5 Results and Performance Analysis

The annual simulation results demonstrate that the designed photovoltaic system achieves strong energy performance under local environmental conditions.

The analysis indicates:

- Stable annual generation performance
- Effective inverter loading
- Acceptable thermal losses
- Efficient utilization of solar resources

Monthly energy production reaches its maximum during summer months due to increased solar irradiation.

The normalized performance analysis shows that total losses remain within acceptable engineering limits.

Overall system behavior confirms that the selected configuration provides reliable and efficient energy production throughout the year.

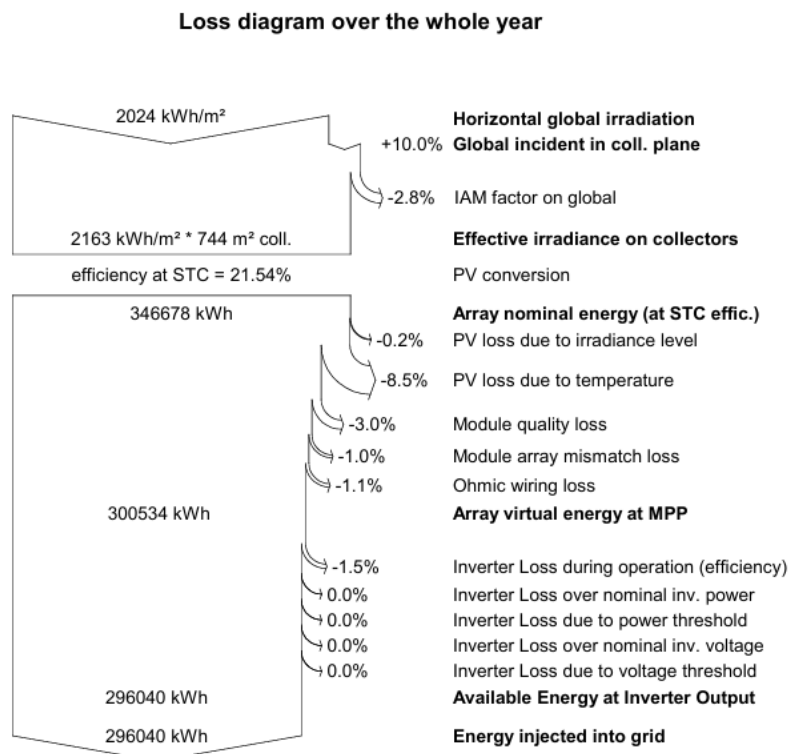


Figure (3.26): Loss Diagram

Single-Line Diagram

The electrical system was represented using a Single-Line Diagram (SLD), showing:

PV Arrays



DC Protection



String Inverters



AC Protection Panel



Main Distribution Board (MDB)

Figure (3.28): Single-Line Diagram

3.5 SketchUp

3.5.1 Introduction to SketchUp

SketchUp was used as a three-dimensional modeling tool to visualize the rooftop installation and evaluate module distribution.

The software was used to:

- Model rooftop dimensions
 - Create mounting structures
 - Arrange PV modules
 - Perform shading assessment
 - Validate installation feasibility
-

3.5.2 Implementation in the Project

Step 1:

Import Site Location

Project coordinates were imported to match actual solar orientation.

Step 2:

Create Rooftop Model

The roof geometry and structural layout were modeled.

Step 3:

Insert PV Modules

PV panels were distributed according to selected electrical configuration.

Step 4:

Shading Analysis

Shading impact was evaluated to minimize energy losses.

Figure (3.29): Roof Modeling

Figure (3.30): PV Layout

Figure (3.31): Shading Analysis

3.5.3 Layout Parameters

Module Orientation:

Landscape

Tilt:

25°

Azimuth:

180° (South)

Inter-row Pitch:

1.8 m

Panel Gap:

0.02 m

Mounting Type:

Fixed Tilt Structure

Maintenance Access:

Included

Figure (3.32): Final SketchUp Layout

3.5.4 Final Rooftop Layout and Shading Results

After completing the rooftop modeling and PV module distribution in SketchUp, a shading assessment was performed to evaluate the effect of surrounding objects and panel spacing on system performance.

The final layout was optimized to maximize energy production while maintaining maintenance accessibility and minimizing mutual shading between rows.

Final Layout Parameters

Module Orientation:

Landscape

Azimuth (Ψ):

180°

Tilt Angle (Σ):

25°

Horizontal Gap:

1.0 m between maintenance paths

Inter-row Pitch:

1.8 m

Space Between Adjacent Panels:

0.02 m

Mounting Configuration:

Single Fixed-Tilt Structure

Stack Configuration:

Single Module Layer

Structure Material:

Galvanized Steel / Aluminum

Installation Type:

Rooftop Mounted

Figure (3.33): Final PV Layout in SketchUp

3.5.5 Shading Analysis Results

A shading analysis was conducted using the developed SketchUp model to identify possible obstacles affecting solar access during different periods of the year.

The simulation considered:

- Roof geometry
- Module spacing
- Solar position throughout the year
- Tilt angle and orientation

The analysis showed that the selected arrangement provides acceptable exposure to solar radiation with minimal shading impact.

Key observations:

1. No significant self-shading was detected between adjacent module rows.
2. Maintenance pathways did not create measurable performance reduction.
3. South-facing orientation improved annual irradiation collection.
4. The selected pitch distance maintained suitable solar access during winter months.
5. The rooftop area was efficiently utilized while preserving structural accessibility.

Figure (3.34): Annual Shading Analysis

3.5.6 Final Layout Performance Assessment

Based on SketchUp visualization and shading evaluation:

Installed DC Capacity:

177.6 kWp

Installed AC Capacity:

150 kW

Total Modules:

320 Modules

Estimated Roof Utilization:

≈ 827 m²

Estimated Structural Load:

≈ 8.8 ton (modules only)

Overall Assessment:

The final rooftop arrangement demonstrates effective use of available installation area with low shading losses and good compatibility with the electrical configuration selected in PVsyst.

The final layout supports reliable annual energy production and complies with practical installation requirements.

Figure (3.35): Final Rooftop PV Arrangement

Figure (3.36): Roof Utilization and Panel Distribution

3.5.7 Integration Between SketchUp and PVsyst

SketchUp was used primarily for:

- Visual layout development
- Panel positioning
- Structural arrangement
- Preliminary shading verification

PVsyst was used for:

- Electrical sizing
- Energy simulation
- Performance evaluation
- Loss analysis
- Annual energy estimation

The combination of both software tools enabled accurate technical and visual validation of the proposed photovoltaic system.

Figure (3.37): Workflow Between SketchUp and PVsyst

